Year 11 Higher

REVISION PACK

Algebra 7

Name: _____

Class: _____

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Worked example: substitution

$s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 5$, $t = 2$ and $a = -3$

Step 1	Put the numbers in	$s = 5 \times 2 + \frac{1}{2} \times (-3) \times 2^2$
Step 2	Work out each part	$= 10 - 6$
Step 3	Answer	$s = 4$

Practice: substitution

1.1 $s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 2$, $t = 3$ and $a = -4$

1.2 $s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 6$, $t = 2$ and $a = -5$

1.3 $s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 4$, $t = 5$ and $a = -1.5$

1.4 $s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 1$, $t = 4$ and $a = 3$

1.5 $s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 10$, $t = 2$ and $a = -2.5$

1.6 $s = ut + \frac{1}{2}at^2$. Work out the value of s when $u = 3$, $t = 6$ and $a = -2$

Worked example: rearranging formulae

Rearrange each formula to make the given letter the subject.

(a) $y = 3x + 2$ Make x the subject.

(b) $E = mc^2$ Make c the subject.

(c) $2x + p = qx + 5$ Make x the subject.

(a)	**Step 1**	Subtract 2	$y - 2 = 3x$
	Step 2	Divide by 3	$x = \dfrac{y-2}{3}$
(b)	**Step 1**	Divide by m	$c^2 = \dfrac{E}{m}$
	Step 2	Square root	$c = \sqrt{\dfrac{E}{m}}$
(c)	**Step 1**	x terms on one side	$2x - qx = 5 - p$
	Step 2	Factorise	$x(2 - q) = 5 - p$
	Step 3	Divide by $(2 - q)$	$x = \dfrac{5-p}{2-q}$

Practice: rearranging formulae

1.7 Rearrange each formula to make the given letter the subject.

(a) $y = 4x + 7$ Make x the subject.

(b) $A = 3r^2$ Make r the subject.

(c) $5x + b = cx + 2$ Make x the subject.

1.8 Rearrange each formula to make the given letter the subject.

(a) $y = 2x - 9$ Make x the subject.

(b) $V = \pi r^2 h$ Make r the subject.

(c) $ax - 3 = 4x + b$ Make x the subject.

1.9 Rearrange each formula to make the given letter the subject.

(a) $y = 6x + 1$ Make x the subject.

(b) $K = \frac{1}{2}mv^2$ Make v the subject.

(c) $7x + p = qx + 3$ Make x the subject.

1.10 Rearrange each formula to make the given letter the subject.

(a) $y = 7x - 2$ Make x the subject.

(b) $P = 5t^2$ Make t the subject.

(c) $px + 5 = 2x + q$ Make x the subject.

1.11 Rearrange each formula to make the given letter the subject.

(a) $y = 3x + 8$ Make x the subject.

(b) $S = 4\pi r^2$ Make r the subject.

(c) $4x - a = bx + 6$ Make x the subject.

1.12 Rearrange each formula to make the given letter the subject.

(a) $y = 9x - 4$ Make x the subject.

(b) $A = 6x^2$ Make x the subject.

(c) $2x + m = nx - 7$ Make x the subject.

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Worked example: changing the subject

$v^2 = u^2 + 2as$. Make s the subject of the formula.

Step 1	Subtract u^2	$v^2 - u^2 = 2as$
Step 2	Divide by $2a$	$s = \dfrac{v^2 - u^2}{2a}$

Practice: changing the subject

1.13 $v^2 = u^2 + 2as$. Make u the subject of the formula.

1.14 $P = 2l + 2w$. Make w the subject of the formula.

1.15 $A = \frac{1}{2}(a + b)h$. Make h the subject of the formula.

1.16 $y = mx + c$. Make m the subject of the formula.

1.17 $V = \frac{1}{3}\pi r^2 h$. Make h the subject of the formula.

1.18 $T = 2a + 3b^2$. Make b the subject of the formula.